

## 06 - Lab Assignment

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Test Case writing, Exercises of Dynamic Testing (Black Box and White Box Test)

Objective 2 Task 1: Reading Activity 2 Task 2: Writing Test Cases. 2 Task 3: Provide softcopy of LabActivity\_day05a for White Box Tests. 3 Task 4: Provide a softcopy of LabActivity\_day05c for Black Box Tests. 6

## Full-Stack QA Automation

### Objective

- Set up a team to experience testing as a collaborative and team effort
- Write proper test cases Functional, Non-Functional, Positive and Negative
- Solve Dynamic Testing exercises

### Task 1: Reading Activity

- Provide reading material:

[https://istqb.org/wp-content/uploads/2024/11/ISTQB\\_CTFL\\_Syllabus\\_v4.0.1.pdf](https://istqb.org/wp-content/uploads/2024/11/ISTQB_CTFL_Syllabus_v4.0.1.pdf)

### Task 2: Writing Test Cases.

Use following description and screen shot to write minimum 10 Test cases in Proper Format also marked Functional, Non-functional, positive and negative cases? identify minimum 10 bugs?

Description: Registration form having form name "Sign Up Form", First Name and Email Address are mandatory fields, email should be in proper format and Name and Last Name should be limited to 20 characters. There would be some checkboxes for allergies and other text box if user don't want to select any allergy. Date Time picker should be in the format "MM/DD/YYYY", Submit will submit the form and Cancel Button will reset all data. (Note: Make sure to focus on test

coverage)

```
<input type="text" id="fname" name="name" required> <input type="text" id="lname" name="lname" required><input type="text" name="email" required>
```

```
<input type="text" type="DatePicker" name=" " > Birthdate
```

```
<input type="checkbox"> Gluten
```

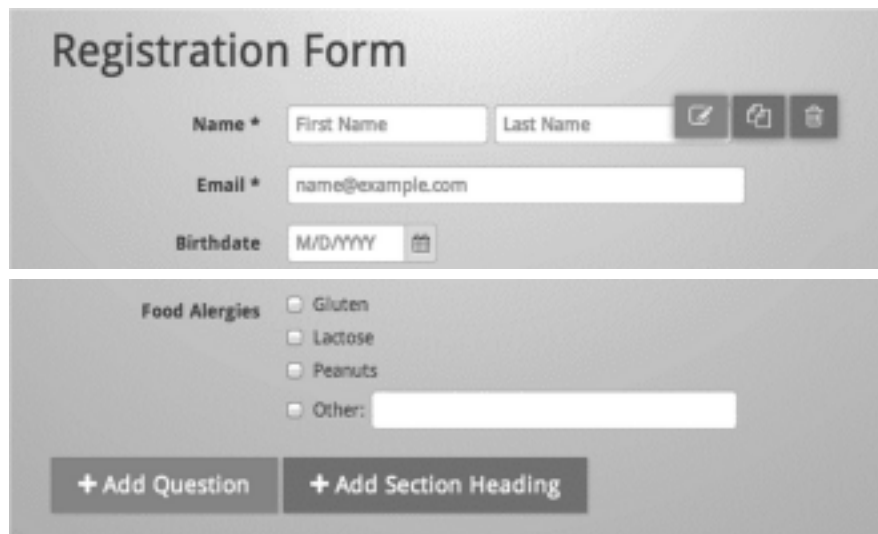
```
<input type="checkbox"> Lactose
```

```
<input type="checkbox"> Peanuts
```

```
<input type="checkbox"> Other:
```

```
<input type="text" name="Alergies">
```

```
<button type="button" class="submitButton">Add Question</button> <button type="submit" class="signupbtn" id="cancelButton">Add Section Heading</button>
```



## Full-Stack QA Automation

### Task 3: White Box Tests.

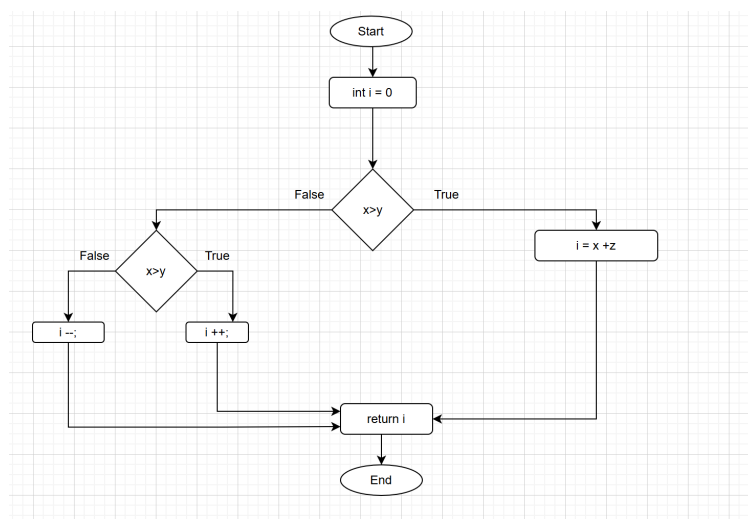
Solve this exercise by utilising knowledge gained in day 05 class.

1. Find 'minimum number of test cases for decision coverage'.
2. Find 'minimum number of test cases for statement coverage'.
3. Find 'minimum number of test cases for condition coverage'.
4. Find number of statements in each code.
5. Find number of conditions in each code.
6. Find number of decisions in each code.

```

int Test_Function(int x, int y, int z)
{
    int i = 0;
    if(x > y)
    {
        i = x + z;
    }
    else
    {
        if (x > y)
        {
            i++;
        }
        else
        {
            i--;
        }
    }
    return i;
}

```

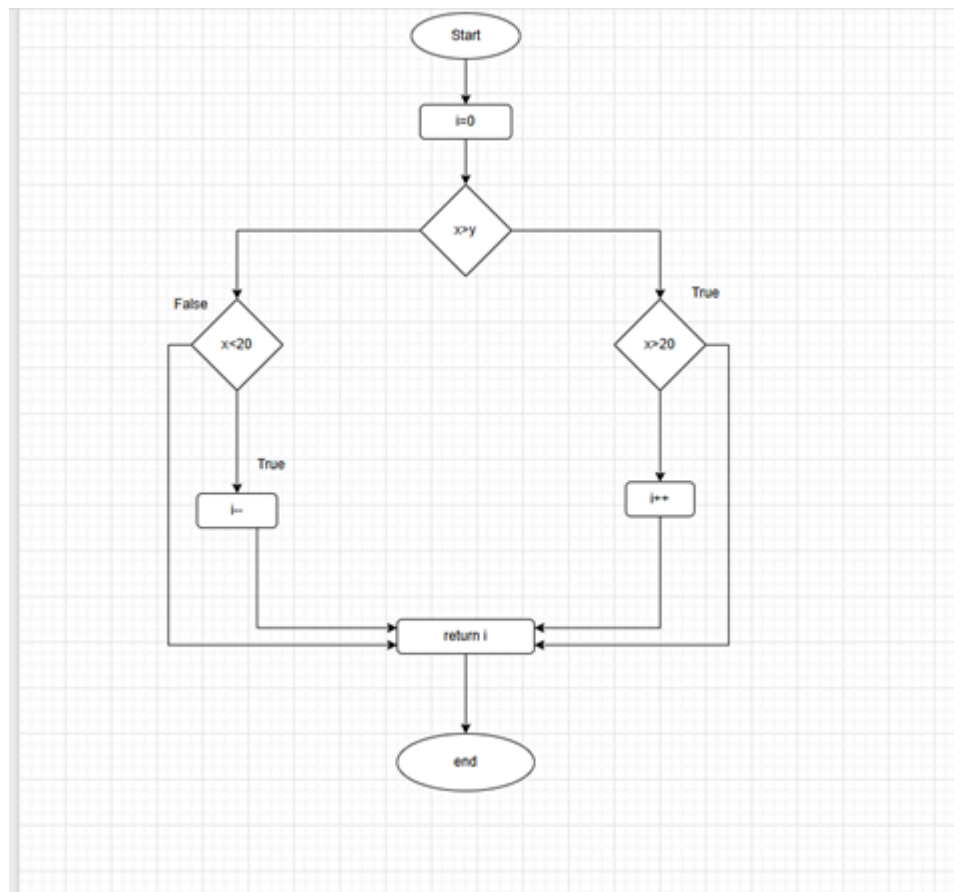


1. Minimum number of test cases for decision coverage is 2.
2. Minimum number of test cases for statement coverage is 3.
3. Minimum number of test cases for condition coverage is 1.
4. The number of statements in code is 2.
5. The number of conditions in code is 2.
6. The number of decisions in code is 2.

```

int Test_Function(int x, int y, int z)
{
    int i = 0;
    if(x > y)
    {
        if (x > 20)
        {
            i++;
        }
    }
    else
    {
        if (x < 20)
        {
            i--;
        }
    }
    return i;
}

```

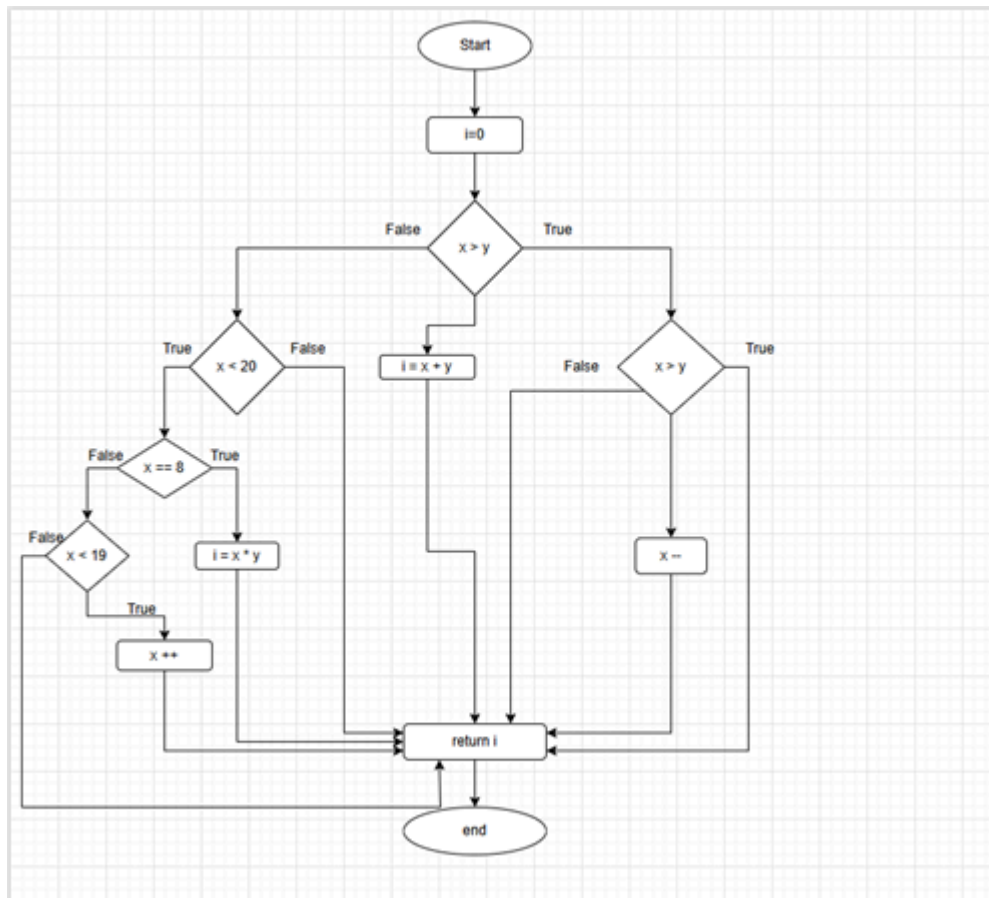


1. Minimum number of test cases for decision coverage is 3.
2. Minimum number of test cases for statement coverage is 2.
3. Minimum number of test cases for condition coverage is 1.
4. The number of statements in code is 2.
5. The number of conditions in code is 4 .
6. The number of decisions in code is 6.

## Full-Stack QA Automation

3

```
int Test_Function(int x, int y, int z)
{
    int i = 0;
    if(x > y)
    {
        while(x > y)
        {
            x--;
        }
        i = x + y;
    }
    else
    {
        if (x < 20)
        {
            if (x == 8)
            {
                i = x * y;
            }
            else
            {
                while(x < 19)
                {
                    x++;
                }
            }
        }
    }
    return i;
}
```



1. Minimum number of test cases for decision coverage is 4.
2. Minimum number of test cases for statement coverage is 4.
3. Minimum number of test cases for condition coverage is 2.
4. The number of statements in code is 4.
5. The number of conditions in code is 5.
6. The number of decisions in code is 6.

```

int Test_Function(int x, int y, int z)
{
    int i = 0;

    switch (x)
    {
        case 0:
            i = x * y;
            break;

        case 1:
            i = x + y;
            break;

        case 2:
            i = x - y;
            break;

        case 3:
            i = x++;
            break;
    }

```

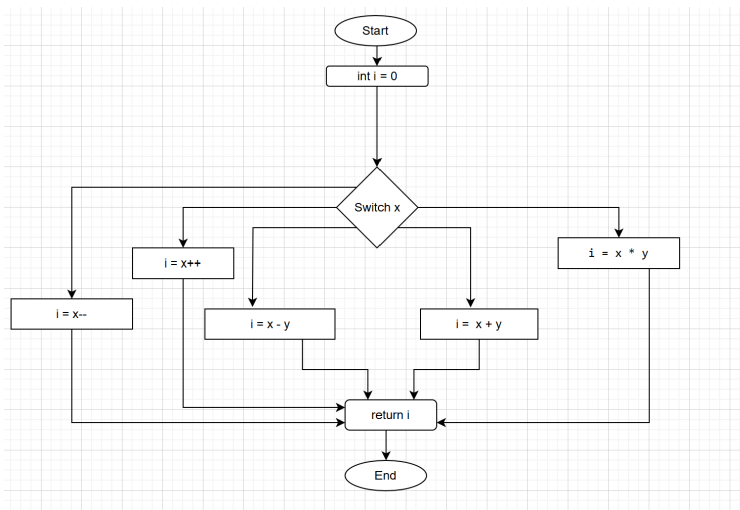
## Full-Stack QA Automation

```

        default:
            i = x--;
            break;
        }

    return i;
}

```



1. Minimum number of test cases for decision coverage is 5.
2. Minimum number of test cases for statement coverage is 5.

3. Minimum number of test cases for condition coverage is 1.
4. The number of statements in code is .
5. The number of conditions in code is .
6. The number of decisions in code is.

5

```
int Test_Function(int x, int y, int z)
{
    int i = 0;

    switch (x)
    {
        case 0:
            i = x * y;
            break;

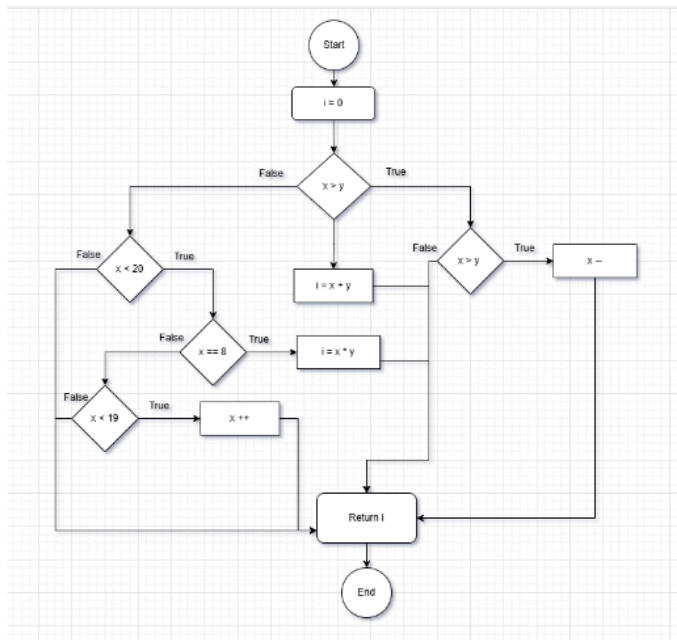
        case 1:
            if(x > y)
            {
                i = z + y;
            }
            break;

        case 2:
            i = x - y;
            break;

        case 3:
            if(z == y)
            {
                i = z + y;
            }
            else
            {
                i = x++;
            }
            break;

        default:
            i = x--;
            break;
    }

    return i;
}
```



1. Minimum number of test cases for decision coverage is 5.
2. Minimum number of test cases for statement coverage is 4.
3. Minimum number of test cases for condition coverage is 2.
4. The number of statements in code is 4.
5. The number of conditions in code is 5.
6. The number of decisions in code is 10.

## Full-Stack QA Automation

### Task 4: Black Box Tests.

Solve this exercise by utilizing knowledge gained in the class.

1. What is the expected result for each of the following test cases?

Rule1 Rule2 Rule3 Rule4

Conditions

Citibank Card Member Yes Yes No No Type of Room Silver Platinum Silver

Platinum Actions

Offer upgrade To Gold Luxury Yes No No No

|                         |     |     |     |    |
|-------------------------|-----|-----|-----|----|
| Offer upgrade to Silver | N/A | Yes | N/A | No |
|-------------------------|-----|-----|-----|----|

- X. Citibank card member, holding a Silver room  
 Y. Non Citibank-member, holding a Platinum room

- A. X – Don't offer any upgrade, Y – Don't offer any upgrade.  
 B. X – Don't offer any upgrade, Y – Offer upgrade to Gold.  
 C. X – Offer upgrade to Silver, Y – Offer upgrade to Silver.  
 D. X – Offer upgrade to Gold, Y – Don't offer any upgrade

**Ans: D is correct option.**

2. Given the following decision table: Which of the following test cases and expected results is VALID?

Rule1 Rule2 Rule3 Rule4

Conditions

Age <21 Years 21-29 Years 30-50 Years >50 Years Insurance Class A A or B B, C or D

C or D Actions

Premium \$100 \$90 \$70 \$70

|        |        |        |       |        |
|--------|--------|--------|-------|--------|
| Excess | \$2500 | \$2500 | \$500 | \$1000 |
|--------|--------|--------|-------|--------|

- A. 23 year old in insurance class A Premium is 100 and excess is,500.
- B. 51 year old in insurance class C Premium is 100 and excess is 00.
- C. 31 year old in insurance class B Premium is 70 and excess is ,500.
- D. 43 year old in insurance class C Premium is 100 and excess is ,5000.

**Ans: C is correct option.**

3. In a system designed to work out the tax to be paid:

Where taxable income does not exceed

Rs600,000 The tax rate is zero

2.5 per cent of the amount exceeding Rs 600,000

Where taxable income exceeds Rs. 600,000 but does not exceed Rs. 1,200,000

Rs15,000 + 12.5 per cent of the amount exceeding Rs1,200,000

Where taxable income exceeds Rs1,200,000 but does not exceed Rs2,400,000

Rs165,000 + 20% of the amount exceeding Rs2,400,000

If taxable income exceeds Rs2,400,000 but does not exceed Rs3,600,000

## Full-Stack QA Automation

Where taxable income exceeds Rs3,600,000 but does not exceed Rs6,000,000

Rs405,000 + 25 per cent of the amount exceeding Rs3,600,000

If taxable income exceeds Rs6,000,000 but does not exceed Rs12,000,000

Rs1,005,000 + 32.5 per cent of the amount exceeding Rs6,000,000

|                                           |                                                                |
|-------------------------------------------|----------------------------------------------------------------|
| Where taxable income exceeds Rs12,000,000 | Rs2,955,000 + 35 per cent of the amount exceeding Rs12,000,000 |
|-------------------------------------------|----------------------------------------------------------------|

List down all VALID and INVALID boundary values and EQUIVALENCE CLASSES for each slab

ANS:

1. **Slab 1: (0 – 600,000)**

- Valid: 0 – 600,000
- Invalid (Lower):  $< 0$
- Invalid (Upper):  $> 600,000$

2. **Slab 2: (600,001 – 1,200,000)**

- Valid: 600,001 – 1,200,000
- Invalid (Lower):  $\leq 600,000$
- Invalid (Upper):  $> 1,200,000$

3. **Slab 3: (1,200,001 – 2,400,000)**

- Valid: 1,200,001 – 2,400,000
- Invalid (Lower):  $\leq 1,200,000$
- Invalid (Upper):  $> 2,400,000$

4. **Slab 4: (2,400,001 – 3,600,000)**

- Valid: 2,400,001 – 3,600,000
- Invalid (Lower):  $\leq 2,400,000$
- Invalid (Upper):  $> 3,600,000$

5. **Slab 5: (3,600,001 – 6,000,000)**

- Valid: 3,600,001 – 6,000,000
- Invalid (Lower):  $\leq 3,600,000$
- Invalid (Upper):  $> 6,000,000$

6. **Slab 6: (6,000,001 – 12,000,000)**

- Valid: 6,000,001 – 12,000,000
- Invalid (Lower):  $\leq 6,000,000$
- Invalid (Upper):  $> 12,000,000$

7. **Slab 7: ( $> 12,000,000$ )**

- Valid: 12,000,001 and onwards
- Invalid (Lower):  $\leq 12,000,000$
- Invalid (Upper): none ( $\infty$ )

## Boundary Value Analysis (BVA)

### 1. Slab 1 (0 – 600,000)

- -1 (Invalid)
- 0 (Valid lower boundary)
- 600,000 (Valid upper boundary)
- 600,001 (Invalid → moves to next slab)

### 2. Slab 2 (600,001 – 1,200,000)

- 600,000 (Invalid → previous slab)
- 600,001 (Valid lower boundary)
- 1,200,000 (Valid upper boundary)
- 1,200,001 (Invalid → moves to next slab)

### 3. Slab 3 (1,200,001 – 2,400,000)

- 1,200,000 (Invalid → previous slab)
- 1,200,001 (Valid lower boundary)
- 2,400,000 (Valid upper boundary)
- 2,400,001 (Invalid → moves to next slab)

### 4. Slab 4 (2,400,001 – 3,600,000)

- 2,400,000 (Invalid)
- 2,400,001 (Valid lower boundary)
- 3,600,000 (Valid upper boundary)
- 3,600,001 (Invalid → next slab)

### 5. Slab 5 (3,600,001 – 6,000,000)

- 3,600,000 (Invalid)
- 3,600,001 (Valid lower boundary)
- 6,000,000 (Valid upper boundary)
- 6,000,001 (Invalid → next slab)

### 6. Slab 6 (6,000,001 – 12,000,000)

- 6,000,000 (Invalid)
- 6,000,001 (Valid lower boundary)
- 12,000,000 (Valid upper boundary)

- 12,000,001 (Invalid → next slab)

#### 7. Slab 7 (> 12,000,000)

- 12,000,000 (Invalid)
- 12,000,001 (Valid lower boundary)
- $\infty$  (Valid upper boundary is not defined, open-ended)
- (No +1 needed beyond  $\infty$ )

4. Let's consider the behaviour of Order Ice Cream Text Box Below. Ice Cream values 1 to 20 is considered valid. A success message is shown. While value 21 to 99 are considered invalid for order and an error message will appear, "Maximum 20 ice cream can be ordered"

|  |   |
|--|---|
|  | t |
|--|---|

Ice Cream: Here is the test condition

|  |       |
|--|-------|
|  | Submi |
|--|-------|

1. Any Number greater than 20 entered in the Ice cream field (let say 21) is considered invalid.
2. Any Number less than 1 that is 0 or below, then it is considered invalid.
3. Numbers 1 to 20 are considered valid
4. Any 3 Digit Number say -100 is invalid.

Design Test that will cover above scenario with all VALID and INVALID boundary values and Equivalence Classes.

#### Equivalence Partitioning (EP):

- Valid: 1 – 20
- Invalid (Lower): -ve – 0
- Invalid (Upper): 21 – onwards

#### Boundary Value Analysis (BVA):

- -1 (Invalid – Lower)
- 1 (Valid – Lower boundary)
- 20 (Valid – Upper boundary)
- 21 (Invalid – Upper boundary)

#### Extra Check (since question mentioned 3-digit numbers):

- 99 (Invalid – Upper, 2-digit edge)

- 100 (Invalid – Upper, 3-digit positive)
- -100 (Invalid – Lower, 3-digit negative)

5. Following password field accepts minimum 8 characters and maximum 15 characters That means results for values in partitions 0-7, 8-15, 16-20 should be equivalent

Enter Password:

Here we will see the Boundary Value Test Cases

| Test Scenario       | Description              | Expected Outcome |
|---------------------|--------------------------|------------------|
| Boundary Value = 0  | System should NOT accept |                  |
| Boundary Value = 8  | System should accept     |                  |
| Boundary Value = 7  |                          |                  |
| Boundary Value = 15 |                          |                  |
| Boundary Value = 16 |                          |                  |
| Boundary Value = 20 |                          |                  |
| Boundary Value = 6  |                          |                  |
| Boundary Value = 14 |                          |                  |
| Boundary Value = 17 |                          |                  |

|                     |  |
|---------------------|--|
| Boundary Value = 19 |  |
|---------------------|--|

## Full-Stack QA Automation

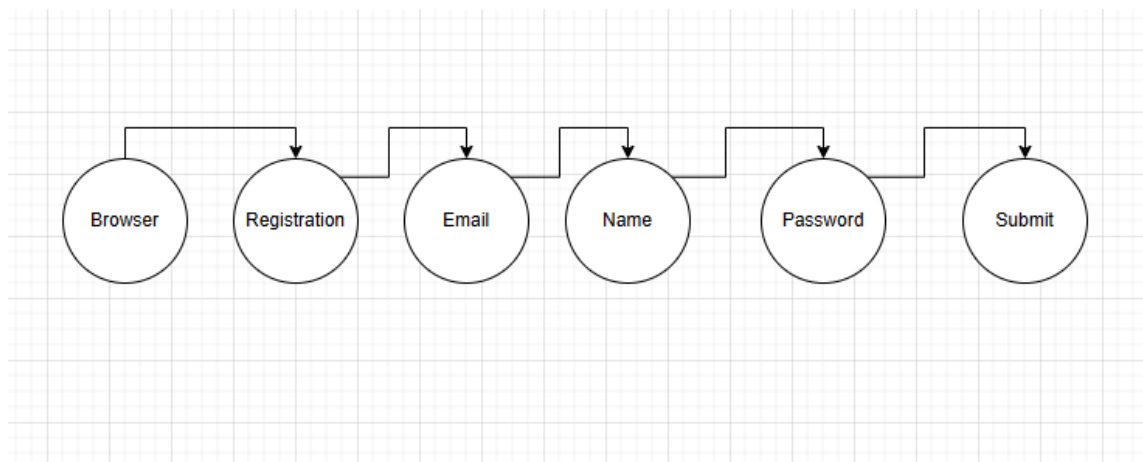
6. Draw Decision Table and state transition diagram of following Registration screen:

### Registration







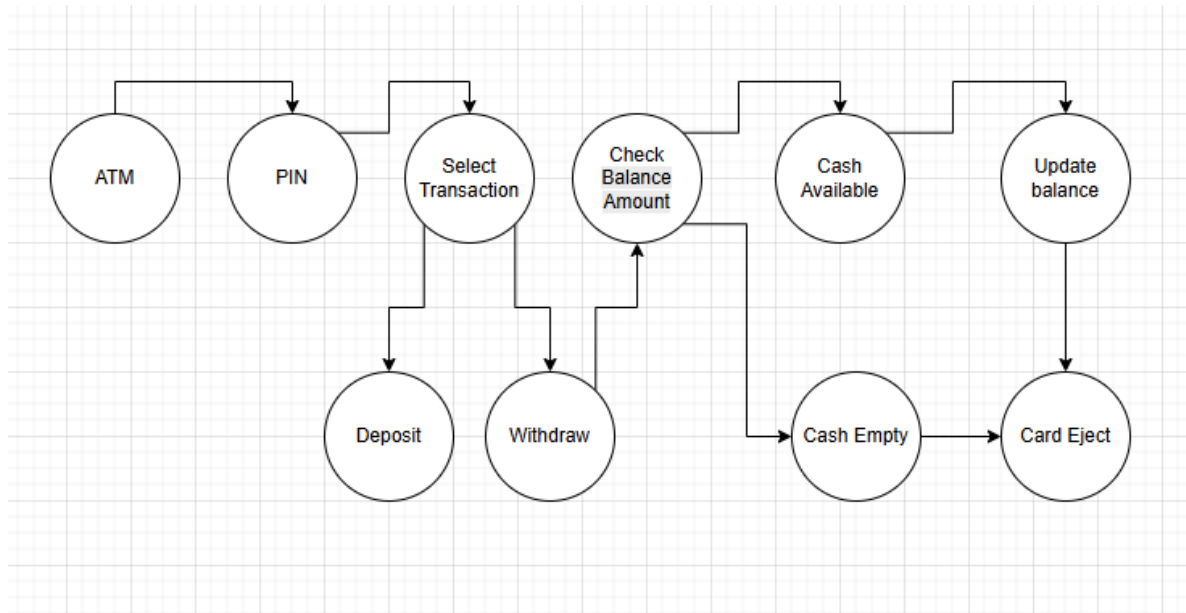


| Conditions | Test Inputs | Test Inputs | Test Inputs | Test Inputs |
|------------|-------------|-------------|-------------|-------------|
| Email      | True        | False       | True        | True        |
| Name       | True        | True        | False       | True        |
| Password   | True        | True        | True        | Fail        |
| Submit     | True        | False       | False       | False       |
| Decision   | Pass        | Fail        | Fail        | Fail        |

7. Draw state transition diagram and decision table of the following scenario: Go to ATM machine to withdraw RS 10,000, requires to the ATM machine and get cash. Later on, go to the ATM and give the same request at same amount but this time ATM refuses the amount because of insufficient balance amount

Note: ATM all default states should also be included in your answer

| Conditions     | Test Inputs | Test Inputs  |
|----------------|-------------|--------------|
| Balance amount | Sufficient  | Insufficient |
| Withdraw       | True        | True         |
| Decision       | Approve     | Reject       |



8. Suppose there is password field which can accept letter from a-z, A-Z and numerical 1-9. Now if we test this password field with straight approach (inefficient one) then we will test this field by entering each alphabet i.e., a to z then A to Z and then numerical 1 to 9 to test whether field accepts them successfully or not. And further we will test it by some invalid values like] (ascii value 93) or @ (ascii value 64).

How many test cases would be there to test? Show all equivalence classes and boundary values

### 1. Test Cases Straight Approach (inefficient)

- Lowercase letters = 26
- Uppercase letters = 26
- Digits (1–9) = 9
- Invalid symbols (jaise @, ], 0, space, etc.) → at least 2–3

Total test cases would be  $26 + 26 + 9 + 3 = 64$  test cases.

### 2. Equivalence Partitioning

- 1 lowercase (e.g., m)
- 1 uppercase (e.g., G)
- 1 digit (e.g., 5)
- 1 invalid symbol (e.g., @)
- 1 invalid zero (0)
- 1 invalid whitespace ( )

Total test cases would be 6 test cases

### 3. Boundary Value Analysis

| Partition | Boundary -1 (Invalid) | Lower Boundary (Valid) | Upper Boundary (Valid) | Boundary +1 (Invalid) |
|-----------|-----------------------|------------------------|------------------------|-----------------------|
| Lowercase | (ASCII 96)            | c                      | z                      | {                     |
| Uppercase | (64)                  | A                      | Z                      | [                     |

|        |      |   |   |   |
|--------|------|---|---|---|
| Digits | (48) | 2 | 4 | : |
|--------|------|---|---|---|

9. Let us consider an example of software application. There is function of software application that accepts only particular number of digits, not even greater or less than that particular number.

Consider an OTP number that contains only 6-digit number, greater and even less than six digits will not be accepted, and the application will redirect customer or user to error page. If password entered by user is less or more than six characters, that equivalence partitioning method will show an invalid OTP. If password entered is exactly six characters, then equivalence partitioning method will show valid OTP.

Provide all equivalence and Boundary values.

**For Equivalent Partition:**

|                                                  |                                                         |                                                            |
|--------------------------------------------------|---------------------------------------------------------|------------------------------------------------------------|
| Length < 6 & any digit<br>e.g. 125 , 8351, 11111 | Length == 6 & any digit<br>e.g. 123456 , 982351, 111111 | Length > 6 & any digit<br>e.g. 12345653 , 9182351, 1111111 |
|--------------------------------------------------|---------------------------------------------------------|------------------------------------------------------------|

**For Boundary Value Analysis:**

|                     |                      |                      |                      |
|---------------------|----------------------|----------------------|----------------------|
| 5 digits e.g. 12365 | 6 digits e.g. 987654 | 6 digits e.g. 123456 | 7 digitse.g. 4321567 |
|---------------------|----------------------|----------------------|----------------------|

10. The next-date problem

Given a day in the format of day-month-year, you need to find the next date for the given date. Perform boundary value analysis and equivalence-class testing for this.

Conditions :

D: 1<Day<31

M: 1<Month<12

Y: 1950 <Year <2025

Example:

Test Case ID Day Month Year Expected Output

|   |   |   |      |          |
|---|---|---|------|----------|
| 1 | 1 | 6 | 2000 | 2-6-2000 |
|---|---|---|------|----------|

**ANS:**

**Day (1–31)**

**Lower boundary:** 1, 2

**Upper boundary:** 30, 31, 32

**Month (1–12)**

**Lower boundary:** 1, 2

**Upper boundary:** 11, 12, 13

**Year (1950–2025)**

**Lower boundary:** 1950, 1951

**Upper boundary:** 2024, 2025, 2026

### Equivalence Classes (ECT)

**Valid Day Class:** 1–28/29/30/31 depending on month

**Invalid Day Class:**  $\leq 0$ ,  $> 31$

**Valid Month Class:** 1–12

**Invalid Month Class:**  $\leq 0$ ,  $> 12$

**Valid Year Class:** 1950–2025

**Invalid Year Class:**  $< 1950$ ,  $> 2025$

| Test Case ID | Day | Month | Year | Expected Output                                  |
|--------------|-----|-------|------|--------------------------------------------------|
| 1            | 1   | 6     | 2000 | 2-6-2000 (normal case)                           |
| 2            | 31  | 1     | 2000 | 1-2-2000 (month rollover)                        |
| 3            | 28  | 2     | 2001 | 1-3-2001 (non-leap year Feb)                     |
| 4            | 29  | 2     | 2000 | 1-3-2000 (leap year Feb)                         |
| 5            | 31  | 12    | 2024 | 1-1-2025 (year rollover)1-1-2025 (year rollover) |
| 6            | 0   | 5     | 2000 | Invalid Input (Day $< 1$ )                       |
| 7            | 32  | 7     | 2000 | Invalid Input (Day $> 31$ )                      |
| 8            | 15  | 0     | 2000 | Invalid Input (Month $< 1$ )                     |
| 9            | 20  | 13    | 2000 | Invalid Input (Month $> 12$ )                    |
| 10           | 10  | 10    | 1949 | Invalid Input (Year $< 1950$ )                   |
| 11           | 25  | 8     | 2026 | Invalid Input (Year $> 2025$ )                   |

## Boundary Value Analysis & Equivalence Partitions Questions

## 1. Flight Booking Age & Class Restriction (Multi-Condition)

Requirement:

- ✎ Economy: Age 2–80
- ✎ Business: Age 18–80
- ✎ First Class: Age 25–65

Task: Identify BVA test cases for each class considering shared and unique boundaries.

Ans:

BVA for Economy Class:

|   |   |    |    |
|---|---|----|----|
| 1 | 2 | 80 | 81 |
|---|---|----|----|

BVA for Business Class:

|    |    |    |    |
|----|----|----|----|
| 17 | 18 | 80 | 81 |
|----|----|----|----|

BVA for First Class:

|    |    |    |    |
|----|----|----|----|
| 24 | 25 | 65 | 66 |
|----|----|----|----|

## 2. Online Exam Time & Score Rule (Dependent Variables)

Requirement:

- ✎ Exam duration: 30–180 minutes.
- ✎ Score is valid only if exam duration  $\geq 45$  minutes.

Task: Create boundary test cases that ensure coverage of both duration range and score validity condition.

Ans:

BVA for Duration Range:

| Lower Limit -1 | Lower Limit | Upper Limit | Upper Limit +1 |
|----------------|-------------|-------------|----------------|
| 29 minutes     | 30 minutes  | 180 minutes | 181 minutes    |

BVA for Score Validity Condition:

| Lower Limit -1 | Lower Limit | Upper Limit |
|----------------|-------------|-------------|
| 44 minutes     | 45 minutes  | 46 minutes  |

| Test Case | Exam duration minutes | Within range (30 -180 minutes) | Score valid $\geq 45$ minutes |
|-----------|-----------------------|--------------------------------|-------------------------------|
| TC_01     | 29                    | invalid                        | invalid                       |
| TC_02     | 30                    | Lower boundary (valid)         | invalid                       |
| TC_03     | 44                    | valid                          | invalid                       |
| TC_04     | 45                    | valid                          | valid                         |
| TC_05     | 46                    | valid                          | valid                         |
| TC_06     | 180                   | Upper boundary(valid)          | valid                         |
| TC_07     | 181                   | invalid                        | invalid                       |

### 3. Car Loan Eligibility (Composite Numeric & Conditional)

Requirement:

- Age: 21–60
- Salary:  $\geq 30,000$  per month
- Loan Amount:  $\leq 20 \times$  monthly salary

Task: Identify BVA test cases for each condition and ECP partitions for combined rules.

BVA for Age:

| Lower Limit -1 | Lower Limit | Upper Limit | Upper Limit +1 |
|----------------|-------------|-------------|----------------|
| 21             | 20          | 60          | 61             |

BVA for Salary:

| Lower Limit -1 | Lower Limit | Just Valid |
|----------------|-------------|------------|
| 29,999         | 30,000      | 35,000     |

BVA for Loan Amount (for salary = 30,000  $\rightarrow$  max = 600,000)

| Lower Limit -1 | Lower Limit | Upper Limit | Upper Limit +1 |
|----------------|-------------|-------------|----------------|
| 59,999         | 600,000     | 600,001     | –              |

| TC_ID | Age | Salary | Loan Amount | Valid/Invalid | Reason             |
|-------|-----|--------|-------------|---------------|--------------------|
| TC_01 | 20  | 30,000 | 200,000     | Invalid       | Age < 21           |
| TC_02 | 21  | 29,000 | 200,000     | Invalid       | Salary < 30k       |
| TC_03 | 21  | 30,000 | 599,999     | Valid         | Lower Boundary     |
| TC_04 | 21  | 30,000 | 600,000     | Valid         | Upper Boundary     |
| TC_05 | 21  | 30,000 | 600,001     | Invalid       | Loan > 20x Salary  |
| TC_06 | 60  | 35,000 | 500,000     | Valid         | Age upper Boundary |
| TC_07 | 61  | 35,000 | 500,000     | Invalid       | Age > 60           |

### 4. Hotel Reservation Check-in & Stay Duration (Date & Range)

Requirement:

- Check-in date: at least 1 day from today, up to 180 days ahead.
- Stay duration: 1–30 days.
- Check-out cannot fall on 25th December (system maintenance).

Task: Design boundary and invalid class test cases.

BVA for Check-in-Date:

| Lower Limit -1 | Lower Limit | Upper Limit | Upper Limit +1 |
|----------------|-------------|-------------|----------------|
| 0 day          | 1 Day       | 180 Days    | 181 Days       |

BVA for Stay Duration:

| Lower Limit -1 | Lower Limit | Upper Limit | Upper Limit +1 |
|----------------|-------------|-------------|----------------|
| 0 day          | 1 Day       | 30 Days     | 31 Days        |

| TC_ID | Check in( days from today) | Stay Duration | Expected Result | Reason                     |
|-------|----------------------------|---------------|-----------------|----------------------------|
| TC_01 | 0                          | 5             | Invalid         | Check-in today not allowed |
| TC_02 | 1                          | 5             | Valid           | Lower Boundary Check in    |
| TC_03 | 180                        | 5             | Valid           | Upper Boundary Check in    |
| TC_04 | 181                        | 5             | Invalid         | Check in > 180             |
| TC_05 | 10                         | 0             | Invalid         | Stay < 1                   |
| TC_06 | 10                         | 1             | Valid           | Lower Boundary stay        |
| TC_07 | 10                         | 30            | Valid           | Upper Boundary Stay        |
| TC_08 | 10                         | 31            | Invalid         | Stay > 30                  |
| TC_09 | 170                        | 15            | Invalid         | Check out falls on 25 Dec  |

## 5. Multi-Stage Discount Eligibility (Tiered Ranges)

Requirement:

- Spend < \$100 → no discount
- Spend \$100–\$499 → 5% discount
- Spend \$500–\$999 → 10% discount
- Spend ≥ \$1,000 → 20% discount

Task: Identify ECP classes and boundary values for each tier.

### Equivalence Class Partitioning (ECP):

| Test Cases | Spend (\$) | Expected Discount | Reason                  |
|------------|------------|-------------------|-------------------------|
| TC_01      | 99         | 0%                | Lower Boundary          |
| TC_02      | 100        | 5%                | Start of 5% tier        |
| TC_03      | 499        | 5%                | Upper Boundary 5% tier  |
| TC_04      | 500        | 10%               | Start of 10% tier       |
| TC_05      | 999        | 10%               | Upper Boundary of 10%   |
| TC_06      | 1000       | 20%               | Start of 20%            |
| TC_07      | 1500       | 20%               | Valid Higher class test |

### 6. Train Ticket Booking Rules (Mixed Fields)

Requirement:

- Max passengers per booking: 1–6.
- If any passenger is under 5 years old, total passengers cannot exceed 4. Task:  
Create BVA cases for passenger count, including dependent age rule.

| Test Cases | Spend (\$) | Expected Discount | Reason  |
|------------|------------|-------------------|---------|
| TC_01      | 0          | No                | Invalid |
| TC_02      | 1          | No                | Valid   |
| TC_03      | 6          | No                | Valid   |
| TC_04      | 7          | No                | Invalid |
| TC_05      | 4          | Yes               | Valid   |
| TC_06      | 5          | Yes               | Invalid |
| TC_07      | 2          | Yes               | Valid   |

### 7. Multi-Field Form Validation (String + Numeric)

Requirement:

- Name: 3–50 characters
- Age: 18–65
- Email: must contain “@” and a domain  
Task: Identify BVA for numeric and string length fields, plus ECP for email validation.

BVA for Name

| Below Minimum | Minimum | Maximum | Above Maximum |
|---------------|---------|---------|---------------|
| 2             | 3       | 50      | 51            |

BVA for Age

| Below Min | Min | Max | Above Max |
|-----------|-----|-----|-----------|
| 17        | 18  | 65  | 66        |

ECP for Email

| Valid                   | Invalid     | Invalid        | Invalid               |
|-------------------------|-------------|----------------|-----------------------|
| Contains "@" and domain | Missing "@" | Missing domain | Invalid domain format |

| Test Case ID | Field | Input Example      | Expected Result |
|--------------|-------|--------------------|-----------------|
| TC_01        | Name  | "Al" (2 chars)     | Invalid         |
| TC_02        | Name  | "Ali" (3 chars)    | Valid           |
| TC_03        | Name  | 50-char string     | Valid           |
| TC_04        | Name  | 51-char string     | Invalid         |
| TC_05        | Age   | 17                 | Invalid         |
| TC_06        | Age   | 18                 | Valid           |
| TC_07        | Age   | 65                 | Valid           |
| TC_08        | Age   | 66                 | Invalid         |
| TC_09        | Email | "user@example.com" | Valid           |
| TC_10        | Email | "userexample.com"  | Invalid         |
| TC_11        | Email | "user@"            | Invalid         |
| TC_12        | Email | "user@domain"      | Invalid         |

## 8. Stock Trading App Order Limits (Conditional)

Requirement:

- Minimum order: 10 shares
  - Maximum: 5,000 shares
  - If price per share > \$500, maximum order is 500 shares.
- Task: Design BVA test cases for both conditions.

BVA for orders

| Minimum +1 | Minimum | Minimum | Maximum -1 |
|------------|---------|---------|------------|
| 09         | 10      | 5000    | 5001       |

BVA for share quantity when price > \$500

| Minimum +1 | Minimum | Minimum | Maximum -1 |
|------------|---------|---------|------------|
| 0          | 1       | 500     | 501        |

| Test Case ID | Price | Order | Expected Result | Reason                   |
|--------------|-------|-------|-----------------|--------------------------|
| TC_01        | 499   | 5000  | Valid           | Normal case              |
| TC_02        | 499   | 5001  | Invalid         | Exceed max 5000          |
| TC_03        | 501   | 500   | Valid           | Normal case              |
| TC_04        | 501   | 501   | Invalid         | Exceed 500 at high price |
| TC_05        | 500   | 5000  | Valid           | Normal case              |
| TC_06        | 500   | 5001  | Invalid         | Exceed max 5000          |
| TC_07        | 10    | 9     | Invalid         | Below min 10             |
| TC_08        | 10    | 10    | Valid           | Normal min case          |

#### 9. University Admission Scoring (Multiple Criteria)

Requirement:

- Academic score: 0–100 (Pass if  $\geq 60$ )
- Entrance test score: 0–200 (Pass if  $\geq 120$ )
- Eligible only if both conditions are met.

Task: Identify BVA cases to test individual boundaries and combined pass/fail cases.

BVA for Academic score

| Minimum -1 | Minimum | Maximum | maximum +1 |
|------------|---------|---------|------------|
| -1         | 0       | 100     | 101        |

BVA for Entrance Test Score

| Minimum -1 | Minimum | Maximum | maximum +1 |
|------------|---------|---------|------------|
| -1         | 0       | 200     | 201        |

| Test Case ID | Academic Score | Entrance Score | Pass/Fail | Reason                                 |
|--------------|----------------|----------------|-----------|----------------------------------------|
| TC_01        | 0              | 200            | Fail      | Academic Score does not meet condition |
| TC_02        | 0              | 201            | Fail      | Both does not meet condition           |
| TC_03        | 100            | 200            | Pass      | Both conditions pass                   |
| TC_04        | 100            | 201            | Fail      | Entrance score condition does not meet |

#### 10. Internet Data Plan Purchase (Tiered + Date Restriction)

Requirement:

- Data amount: 1–100 GB
  - Special promo: For purchases in January, max limit increases to 150 GB.
- Task: Identify BVA and ECP test cases for normal and promo months.

BVA for Data amount

| Minimum -1 | Minimum | Maximum | maximum +1 |
|------------|---------|---------|------------|
| 0          | 1       | 100     | 101        |

BVA for Data Amount in January

| Minimum -1 | Minimum | Maximum | maximum +1 |
|------------|---------|---------|------------|
| 0          | 1       | 150     | 151        |

ECP for Data amount

| Data Amount | Month       | Expected Result |
|-------------|-------------|-----------------|
| 150         | December 30 | Invalid         |
| 150         | January 1   | Valid           |
| 150         | February 28 | Invalid         |

| Test Case ID | Data Amount | Month-Date | Expected Result | Reason       |
|--------------|-------------|------------|-----------------|--------------|
| TC_01        | 100         | April 15   | Valid           | Normal Month |

|       |     |              |         |                               |
|-------|-----|--------------|---------|-------------------------------|
| TC_02 | 100 | January 2    | Valid   | Normal Amount                 |
| TC_03 | 101 | September 17 | Invalid | Exceeds Data amount           |
| TC_04 | 101 | January 14   | Valid   | Boundary Month true condition |
| TC_05 | 150 | July 9       | Invalid | Exceeds Data amount           |
| TC_06 | 150 | January 4    | Valid   | Boundary Month                |
| TC_07 | 151 | April 13     | Invalid | Exceeds Data amount           |
| TC_08 | 151 | January 1    | Invalid | Exceeds Data amount           |

## Full-Stack QA Automation

### Decision Table Problems

1. Online Loan Approval (multiple conditions & priorities)

Conditions: Applicant Age ( $\geq 21$ ), Employment Status (Employed/Unemployed), Monthly Salary ( $\geq 50,000$ ), Credit Score ( $\geq 700$ ), Existing Loan (Yes/No). Actions: Approve Full, Approve Partial (50% requested), Reject, Request Manual Review. Task: Create a complete decision table (including impossible/contradictory combinations), mark rule priority (e.g., manual review overrides auto-approve), and derive minimum set of test cases that cover all unique outcomes.

Decision Table:

| Rule | Age $\geq 21$ | Employment | Salary $\geq 50,000$  | Credit $\geq 700$ | Existing Loan | Decision      |
|------|---------------|------------|-----------------------|-------------------|---------------|---------------|
| R1   | No            | Any        | Any                   | any               | any           | reject        |
| R2   | Yes           | Unemployed | Any                   | any               | any           | reject        |
| R3   | Yes           | Employed   | No <50K               | any               | any           | reject        |
| R4   | Yes           | Employed   | Yes                   | No <700           | any           | reject        |
| R5   | Yes           | Employed   | Yes                   | yes               | no            | approve       |
| R6   | Yes           | Employed   | Yes                   | yes               | yes           | approve       |
| R7   | Yes           | Employed   | salary=50k,credit=700 | any               | any           | Manual review |
| R8   | Yes           | Employed   | Yes                   | yes               | any           | manual        |
| R9   | Yes           | Employed   | yes                   | yes               | any           | manual        |

## Test Cases

| Test Case Id | Age ≥21 | Employment | Salary ≥50,000 | Credit ≥700 | Existing Loan | Decision      |
|--------------|---------|------------|----------------|-------------|---------------|---------------|
| TC1          | 18      | Employed   | 60k            | 720         | no            | reject        |
| TC2          | 30      | Unemployed | 0              | 650         | no            | reject        |
| TC3          | 35      | Employed   | 40K            | 750         | no            | reject        |
| TC4          | 40      | Employed   | 60k            | 650         | no            | reject        |
| TC5          | 28      | Employed   | 55k            | 750         | no            | approve       |
| TC6          | 32      | Employed   | 70k            | 780         | yes           | approve       |
| TC7          | 29      | Employed   | 50k            | 700         | no            | Manual review |

## 2. E-commerce Return Policy (time + item type + condition)

Conditions: Days since delivery ( $\leq 30$ , 31–90,  $> 90$ ), Item Type (Electronics/Clothing/Perishable), Opened (Yes/No), Proof of Purchase (Yes/No).

Actions: Full Refund, Exchange Only, No Return, Refund with Restocking Fee.

Task: Build the decision table, indicate which combinations are valid, and produce test cases that exercise all return outcomes including special-case exceptions (e.g., perishable items).

Decision Table:

| Rule | Days Since Delivery | Item type   | Opened | Purchased | Action                     |
|------|---------------------|-------------|--------|-----------|----------------------------|
| R1   | $\leq 30$           | Electronics | No     | Yes       | Full Refund                |
| R2   | $\leq 30$           | Electronics | Yes    | Yes       | Refund with Restocking fee |
| R3   | $\leq 30$           | Clothing    | Any    | Yes       | Full refund                |
| R4   | $\leq 30$           | Perishable  | Any    | Any       | No return                  |
| R5   | 31–90               | Electronics | Any    | Yes       | Exchange only              |
| R6   | 31–90               | Clothing    | Any    | Yes       | Exchange only              |
| R7   | 31–90               | Perishable  | Any    | Any       | No return                  |
| R8   | $> 90$              | Any item    | Any    | Any       | No return                  |
| R9   | Any days            | Any item    | any    | no        | No return                  |

Test Cases:

| Test Cases ID | Days | Item type   | Opened | Purchased | Expected Result            |
|---------------|------|-------------|--------|-----------|----------------------------|
| TC1           | 15   | Electronics | No     | Yes       | Full Refund                |
| TC2           | 21   | Electronics | Yes    | Yes       | Refund with Restocking fee |
| TC3           | 9    | Clothing    | Yes    | Yes       | Full refund                |
| TC4           | 6    | Perishable  | No     | Yes       | No return                  |
| TC5           | 47   | Electronics | No     | Yes       | Exchange only              |
| TC6           | 60   | Clothing    | Yes    | Yes       | Exchange only              |
| TC7           | 80   | Perishable  | No     | Yes       | No return                  |
| TC8           | 125  | Clothing    | No     | Yes       | No return                  |
| TC9           | 24   | Electronics | No     | No        | No return                  |

3. Access Control for Document Management (role + clearance + doc classification)

Conditions: User Role (Viewer/Editor/Admin), Security Clearance Level (Low/Medium/High), Document Classification (Public/Internal/Confidential/Secret), Access Type (Read/Write/Delete).

Actions: Grant Access, Deny Access, Grant Read-only, Escalate to Admin. Task: Construct the decision table and identify rules that produce conflicting grants; resolve conflicts with a stated policy and derive test cases.

| Rule | User Role | Clearance    | Doc Classification           | Access type  | Action             |
|------|-----------|--------------|------------------------------|--------------|--------------------|
| R1   | Viewer    | Any          | Public                       | Read         | Grant Access       |
| R2   | Viewer    | Any          | Public                       | Write/Delete | Deny Access        |
| R3   | Viewer    | Low/Med/High | Internal/Confidential/Secret | Any          | Grant Read only    |
| R4   | Editor    | Medium/High  | Internal/Confidential        | Read/Write   | Grant Access       |
| R5   | Editor    | Medium/High  | Secret                       | Read         | Grant Read only    |
| R6   | Editor    | Medium/High  | Secret                       | Write/Delete | Escalate to access |
| R7   | Editor    | Low          | Internal/Confidential/Secret | Any          | Deny Access        |

|     |                              |                              |        |        |                   |
|-----|------------------------------|------------------------------|--------|--------|-------------------|
| R8  | Admin                        | Any                          | Any    | Any    | Grant Access(All) |
| R9  | Any Role                     | Clearance < doc for required | Any    | Any    | Deny Access       |
| R10 | Any Role (except Admin Role) | High                         | Secret | Delete | Escalata to Admin |

### Conflict Rules:

Conflict A: Editor with High Clearance on Secret Document – Read.

- R5 says → Grant Read-only
- R8 (Admin override) does not apply (user is not admin), so R5 is valid.  
Resolved by priority policy: "More restrictive rule wins (least privilege principle)"

Conflict B: Editor with Medium Clearance on Secret Document – Write.

- R6 → Escalate
- R9 → Deny (since clearance may be below "Secret")  
Resolution: Define required clearance levels (e.g., Secret requires High). Thus R9 triggers → Deny (clearance check first).

### Conflict Resolution Policy:

- Policy: Always check clearance vs classification first. If clearance insufficient → Deny.
- If clearance sufficient → apply role permissions.
- If overlap remains → choose least privilege (Grant Read-only instead of full Grant).
- Admin overrides all.

### Derived Test Cases

| Test Case ID | User Role | Clearance | Document Classification | Access Type | Expected Result |
|--------------|-----------|-----------|-------------------------|-------------|-----------------|
| TC_01        | Viewer    | Low       | Public                  | Read        | Grant Access    |
| TC_02        | Viewer    | High      | Confidential            | Write       | Grant Read-only |
| TC_03        | Viewer    | Medium    | Secret                  | Delete      | Deny Access     |

|       |        |        |              |        |                   |
|-------|--------|--------|--------------|--------|-------------------|
| TC_04 | Editor | Medium | Internal     | Write  | Grant Access      |
| TC_05 | Editor | Low    | Confidential | Read   | Deny Access       |
| TC_06 | Editor | High   | Secret       | Read   | Deny Access       |
| TC_07 | Editor | Medium | Secret       | Write  | Deny Access       |
| TC_08 | Editor | High   | Secret       | Delete | Escalate to Admin |
| TC_09 | Admin  | Low    | Secret       | Delete | Grant Access      |
| TC_10 | Viewer | Any    | Public       | Delete | Deny Access       |

#### 4. Mobile App Notification Rules (user prefs + global status)

Conditions: User Preference (All/Important/None), Do-Not-Disturb (On/Off), App Critical Alert (Yes/No), Battery Saver Mode (On/Off).

Actions: Send Push, Send Silent, Suppress, Queue for Later.

Task: Create the decision table including precedence (critical alerts bypass some settings), and list test cases for boundary/edge rules (e.g., critical alert + DND + battery saver).

Decision Table

| Rule | User Pref | DND | Critical Alert | Battery Saver | Action          |
|------|-----------|-----|----------------|---------------|-----------------|
| R1   | All       | Off | No             | Off           | Send Push       |
| R2   | All       | Off | No             | On            | Queue for Later |
| R3   | All       | On  | No             | Any           | Suppress        |
| R4   | Important | Off | No             | Off           | Send silent     |
| R5   | Important | Off | No             | On            | Queue for later |
| R6   | Important | On  | No             | Any           | Suppress        |
| R7   | None      | Any | No             | Any           | Suppress        |
| R8   | Any       | Any | Yes            | Off           | Send Push       |
| R9   | Any       | Any | Yes            | On            | Send Silent     |

Test Cases

| Test Case ID | User Pref | DND | Critical Alert | Battery Saver | Expected Result |
|--------------|-----------|-----|----------------|---------------|-----------------|
| TC_01        | All       | Off | No             | On            | Send Push       |

|       |           |     |     |     |                 |
|-------|-----------|-----|-----|-----|-----------------|
| TC_02 | All       | Off | No  | Off | Queue for later |
| TC_03 | All       | On  | No  | Off | Suppress        |
| TC_04 | Important | Off | No  | Off | Send Silent     |
| TC_05 | Important | On  | No  | Off | Suppress        |
| TC_06 | None      | Off | No  | Off | Suppress        |
| TC_07 | Any       | On  | No  | Off | Send Push       |
| TC_08 | Any       | On  | Yes | On  | Send Silent     |
| TC_09 | None      | On  | Yes | On  | Send Silent     |
| TC_10 | Important | On  | Yes | Off | Send Push       |

5. Shipping Cost & Method Selection (weight + destination + promo) Conditions: Weight ( $\leq 1\text{kg}$ ,  $>1-5\text{kg}$ ,  $>5\text{kg}$ ), Destination (Local/International), Delivery Speed (Standard/Express), Promo Code (Valid/Invalid).

Actions: Charge Standard Fee, Charge Express Fee, Free Shipping, Reject (cannot ship).

Task: Produce the decision table and derive test cases showing when promo codes override shipping fees and when weight/destination combinations are not shippable.

Decision Table

| Rule | Weight            | Destination   | Speed    | Promo Code | Action              |
|------|-------------------|---------------|----------|------------|---------------------|
| R1   | $\leq 1\text{kg}$ | Local         | Standard | Invalid    | Charge Standard Fee |
| R2   | $\leq 1\text{kg}$ | Local         | Express  | Invalid    | Charge Extend Fee   |
| R3   | $\leq 1\text{kg}$ | International | Standard | Invalid    | Charge Standard fee |
| R4   | $\leq 1\text{kg}$ | International | Express  | Invalid    | Charge Extend fee   |
| R5   | $>1-5\text{kg}$   | Local         | Standard | Invalid    | Charge Standard fee |
| R6   | $>1-5\text{kg}$   | Local         | Express  | Invalid    | Charge Extend fee   |
| R7   | $>1-5\text{kg}$   | International | Standard | Invalid    | Charge Standard fee |
| R8   | $>1-5\text{kg}$   | International | Express  | Invalid    | Charge Extend fee   |

|     |      |               |          |         |                     |
|-----|------|---------------|----------|---------|---------------------|
| R9  | >5kg | Local         | Standard | Invalid | Change Standard fee |
| R10 | >5kg | Local         | Express  | Invalid | Reject              |
| R11 | >5kg | International | Any      | Invalid | Reject              |
| R12 | Any  | Any           | Any      | Valid   | Free Shipping       |

## Test Cases

| Test Case ID | Weight | Destination   | Speed    | Promo Code | Expected Result     |
|--------------|--------|---------------|----------|------------|---------------------|
| TC_01        | 0.5Kg  | Local         | Standard | Invalid    | Charge Standard Fee |
| TC_02        | 0.5Kg  | Local         | Express  | Invalid    | Charge Express Fee  |
| TC_03        | 1Kg    | International | Standard | Invalid    | Charge Standard Fee |
| TC_04        | 4Kg    | International | Express  | Invalid    | Charge Express Fee  |
| TC_05        | 5.1Kg  | Local         | Standard | Invalid    | Charge Standard Fee |
| TC_06        | 5.1Kg  | Local         | Express  | Invalid    | Reject              |
| TC_07        | 6Kg    | International | Any      | Invalid    | Reject              |
| TC_08        | 2Kg    | Local         | Express  | Valid      | Free Shipping       |
| TC_09        | 1Kg    | International | Standard | Valid      | Free Shipping       |
| TC_10        | 6Kg    | International | Express  | Valid      | Reject              |

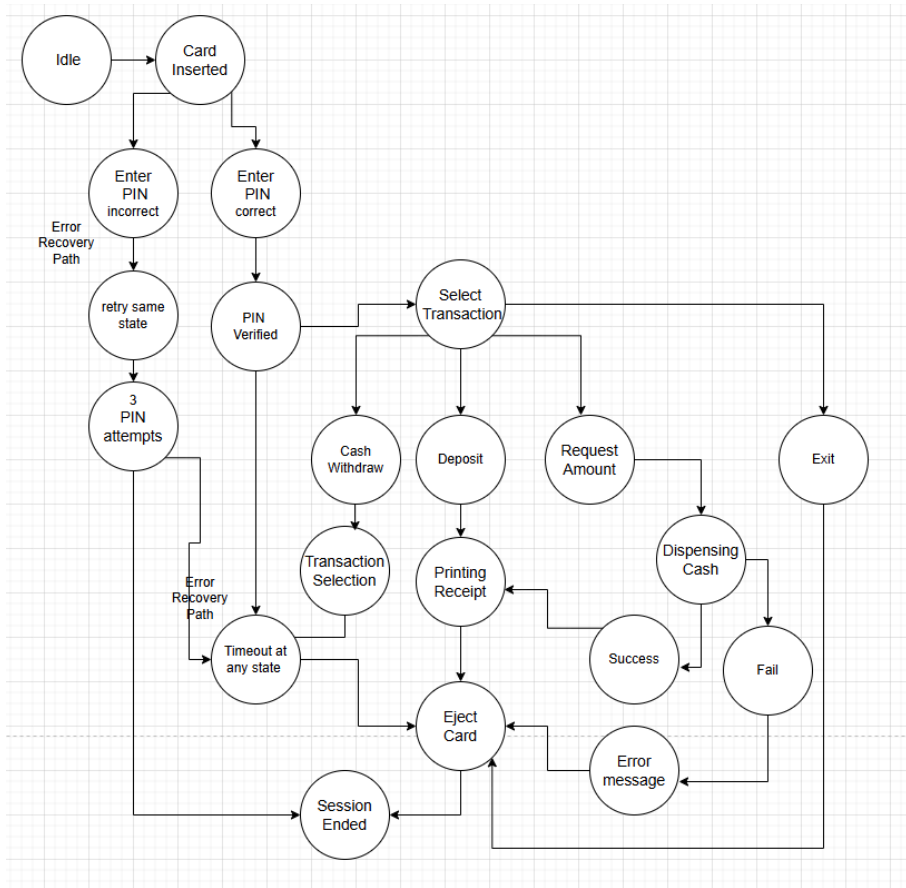
## State-Transition Problems

### 1. ATM Transaction Session

States: Idle, CardInserted, PINVerified, TransactionSelection, DispensingCash, PrintingReceipt, EjectCard, SessionEnded.

Events/Triggers: InsertCard, EnterPIN(correct/incorrect), SelectTransaction(CashWithdraw/Balance/Exit), RequestAmount, DispenseSuccess/Fail, Timeout, Cancel.

Task: Draw the state-transition diagram inc.g., max 3 PIN attempts → retain card), timeouts, and error recovery paths; produce test sequences that cover successful withdrawal, PIN lockout, cancel mid-flow, and cash-dispense failure.

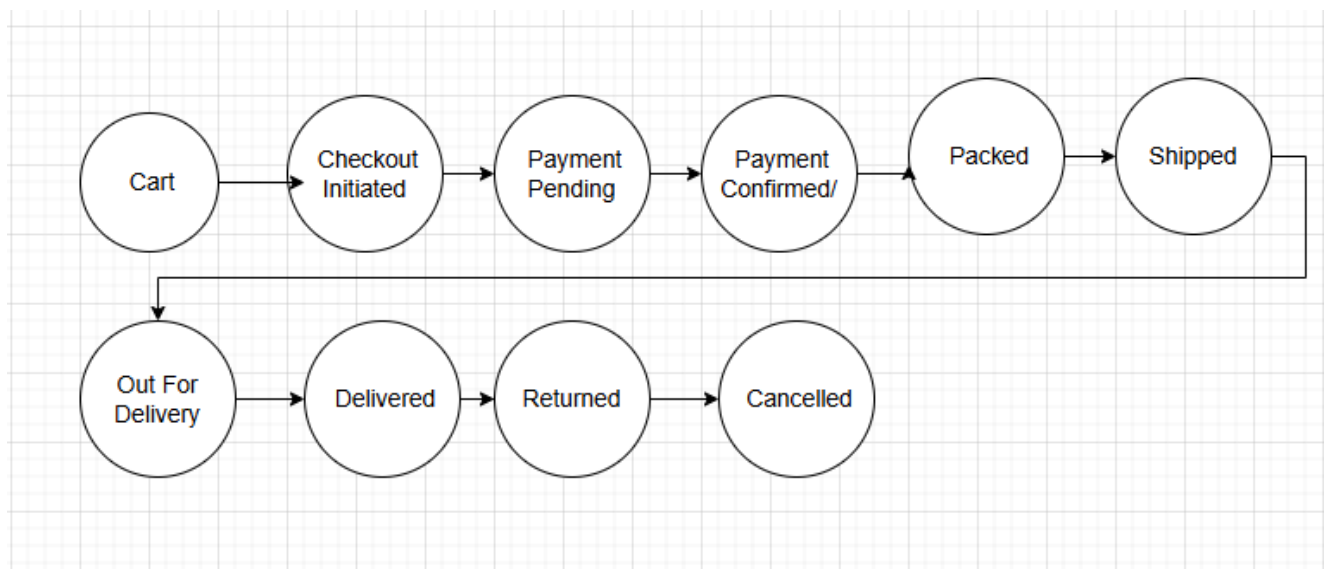


## 2. Online Shopping Order Lifecycle

States: Cart, CheckoutInitiated, PaymentPending, PaymentConfirmed, Packed, Shipped, OutForDelivery, Delivered, Returned, Cancelled.

Events: AddToCart, Checkout, Pay(success/fail), Pack, Ship, Deliver, RequestReturn, ApproveReturn, Cancel.

Task: Model the state machine including allowed transitions (e.g., cancellation allowed only before shipping), exceptional transitions (payment failure → retry or cancel), and

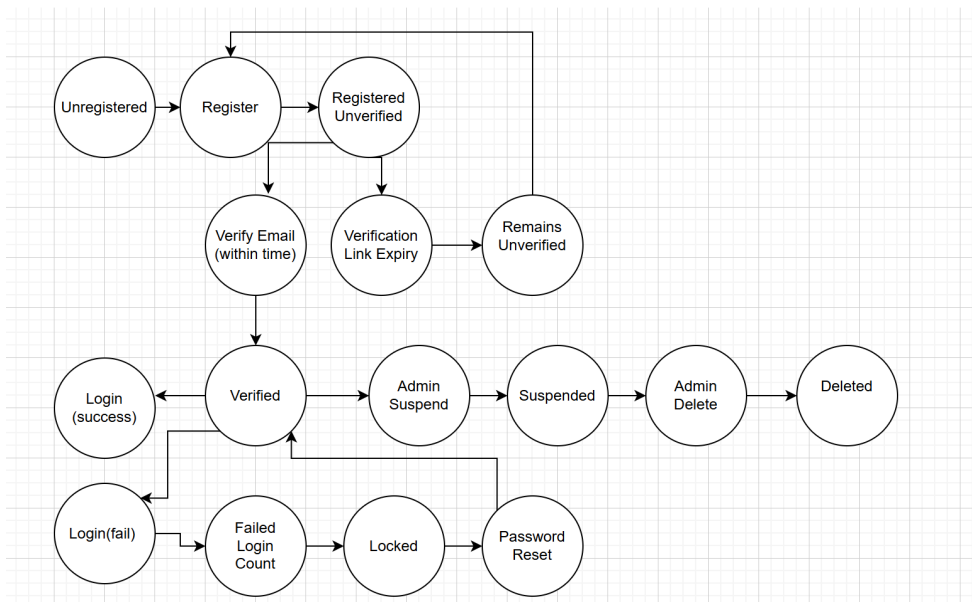


produce test cases for normal flow, payment retry, cancellation attempt after shipping, and return processing. and return processing.

| Test Case                     | Steps                                                                                    | Expected Result                                                  |
|-------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| Normal Flow                   | AddToCart → Checkout → Pay(success) → Pack → Ship → Deliver (OutForDelivery → Delivered) | Order ends in Delivered                                          |
| Payment Retry                 | Checkout → Pay(fail) → PaymentPending → Pay(success) → Pack → Ship → Deliver             | Order Delivered after retry                                      |
| Cancel Attempt After Shipping | Pay(success) → Pack → Ship → Cancel (Attempt)                                            | Reject cancel (disallowed post-ship); order proceeds to delivery |
| Return Processing             | Pay(success) → Pack → Ship → Deliver → RequestReturn → Approve Return                    | Order ends in Returned                                           |

### 3. User Account Lifecycle with Email Verification & Lockout

States: Unregistered, RegisteredUnverified, Verified, Suspended, Locked, Deleted.  
Events: Register, SendVerificationEmail, VerifyEmail, Login(success/fail), FailedLoginCountThreshold, AdminSuspend, AdminDelete, PasswordReset. Task: Create the state transition diagram including timed transitions (e.g., verification link expiry), account lockout after failed attempts, and tests for (a) registering and verifying within time, (b) expired verification link, (c) lockout and password reset path.



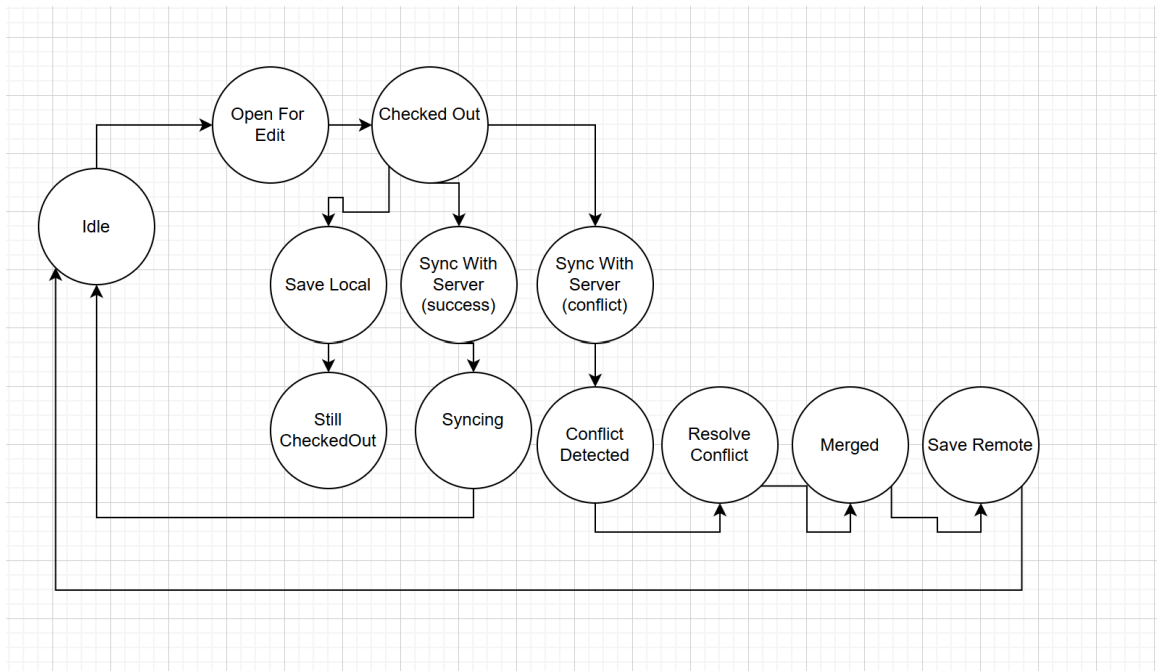
### 4. Document Editing Collaboration (optimistic concurrency)

States per document: Idle, CheckedOut(LocalEdit), Syncing, ConflictDetected, Merged, ReadOnly.

Events: OpenForEdit, SaveLocal, SyncWithServer(success/conflict), ResolveConflict, SaveRemote, RevokeEdit.

Task: Draw transition diagram showing conflict detection when concurrent saves occur, include resolution flows (manual merge, overwrite, abort), and produce test sequences

demonstrating a conflict scenario and successful automatic merge.



#### 5. Ticket Booking with Seat Hold Timeout

States: Browsing, SelectingSeats, SeatsHeld, PaymentProcessing, Confirmed, HoldExpired, Cancelled.

Events: SelectSeat, HoldSeats(timeout), EnterPayment, PaymentSuccess, PaymentFail, Timeout (hold expiry), UserCancel.

Task: Model transitions including hold expiry behavior (e.g., if hold expires during payment, seats released), and create test scenarios: successful booking within hold, payment delay causing hold expiry and seat release, and user cancel during payment.

